VICTOR NAN FERNANDEZ-AYALA

PhD Candidate in Robotics and Control at KTH Royal Institute of Technology. Research in safety-critical control, model-based controllers, RL, multi-robot systems and deploying embodied AI in realistic dynamic environments. Extensive experience bridging learning, control, perception and mechatronics from algorithm design to validation in multiple fields.

CONTACT

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Victor Nan Fernandez-Ayala
Google Scholar

SKILLS

Programming

Python
C/C++
MATLAB/Simulink
LaTeX

ML / Robotics

Reinforcement learning
Robot control & planning
Robotics simulation
Embedded systems

Platforms

ROS1/ROS2
Gazebo, Isaac Sim/Lab, Newton
PX4, ArduPilot
KiCad, SolidWorks, Fusion 360
ESP32, STM32, CAN, SimpleFOC

Research Areas

Safety-critical control
Multi-agent systems
Model-based controllers
Underwater & space robotics
Mobile manipulation
Human-in-the-loop autonomy
Edge deployment of AI and vision models

Languages

English
Spanish
Romanian
Swedish

CERTIFICATES

L1 high-powered rocketry license by NAR
EASA Drone License A2
EASA Drone License A1/A3
Deep Learning Neural Nets specialization by deeplearning.ai (CNNs and RNNs)

PROFILE

PhD candidate in Robotics and Control at KTH and affiliated WASP researcher with a strong focus on getting learning and control methods to work on real systems. Experience spans underwater & space robots, mobile manipulation, multi-robot systems, agriculture, construction and aerial robotics, with work ranging from algorithm design to hardware integration and experiments.

WORK HISTORY

09/2022 – Present
KTH DCS, Stockholm, Sweden
PhD Candidate in Robotics and Control
Research in multi-robot systems, safety-critical control and human-in-the-loop autonomy. Extensive use of vision models and learning-based sysID in underwater (SHARCEX project), agriculture (EU CANOPIES project) and construction (DF Smart Construction project) for both simulation and experiments. Supervision of 9+ master student theses and 15+ ongoing & completed publications.

07/2024 – 07/2025
Animum, Stockholm, Sweden
Co-Founder & CTO
Led the technical roadmap for an autonomous mobile manipulation system for supermarket shelf stocking and fronting. Integrated perception, planning and control and used VLMs for product and scene understanding to guide high-level planning. 2+ patents and & technical demo sold to ICAx.

2025 – Present
Stockholm, Sweden
Technical Advisor and Startup Mentor
Advising two robotics startups on PX4-based drone systems and swarming pipelines for drone firefighting in wildfires, and humanoid-enabled manufacturing using provable-correct visual servoing.

01/2022 – 09/2022
KTH Smart Mobility Lab
Stockholm, Sweden
Research Engineer
Supported robotics experiments and autonomy research for agriculture navigation and grape collection in an italian vineyard, and autonomy experiments for warehouse logistics for Co4Robots.

2019 – 2020
Drone Hopper
Madrid, Spain
Founding Engineer
Built ROS and Gazebo simulation tools, and developed autopilot and control software for heavy-lift electrical and ICE drones by modifying and contributing back to the ArduPilot software stack.

2019
Continental Automotive
Timişoara, Romania
Junior Electronics Intern
Built a capacitance measurement device for an intelligent digital windshield screen, from circuit design and LTspice simulation to PCB design and embedded JavaScript programming.

EDUCATION

2022 – 2026
KTH, Stockholm, Sweden
PhD in Robotics and Control
With all credits, publications and licentiate defense completed. Expected PhD defense fall 2026.

2020 – 2022
KTH, Stockholm, Sweden
MSc in Aerospace Engineering
Specialization in systems and control. Part of the Boomerang and the B2D2 Rexus team where student experiments go to space in the form of CubeSats in collaboration with ESA and DLR. Team leader & founder of KTH ALPHA, a High Altitude Long Endurance UAV built by more than 20 students across different disciplines which led to a publication in the 33rd Congress of the ICAS.

2016 – 2020
UC3M, Madrid, Spain
BSc in Aerospace Engineering
Specializing in aerospace systems and control with multiple honors in related subjects (Control System Analysis and Design, Circuit Design, Programming, etc). Exchange year at **Georgia Institute of Technology** in Atlanta, USA, where I became a member of the student Ramblin' Rocket Club.

SELECTED PUBLICATIONS

Estimating unknown dynamics and cost as a bilinear system with Koopman-based Inverse Optimal Control
V. Nan Fernandez-Ayala, S. A. Deka and D. V. Dimarogonas
2026 IEEE Transactions on Automatic Control (**TAC**), under revision

Marinarium: a New Arena to Bring Maritime Robotics Closer to Shore
I. Torroba, D. Dorner, **V. Nan Fernandez-Ayala** et al.
2026 IEEE Transactions on Field Robotics (**TFR**), under revision

ConstrucTwin: Digital Twin-Driven Multirobot Construction System Toward Industry 5.0
Z. Liu, J. Silva, R. Zhong, Q. Qin, N. Roy, **V. Nan Fernandez-Ayala** et al.
2026 IEEE Transactions on Systems, Man, and Cybernetics: Systems (**TSMC**)

Robust Visual Servoing under Human Supervision for Assembly Tasks
V. Nan Fernandez-Ayala, J. Silva, M. Guo and D. V. Dimarogonas
2025 European Journal of Control (**EJC**)

Efficient Coordination and Synchronization of Multi-Robot Systems Under Recurring Linear Temporal Logic
D. Peron, **V. Nan Fernandez-Ayala**, E. E. Vlahakis and D. V. Dimarogonas
2025 IEEE International Conference on Robotics and Automation (**ICRA**)

From Pixels to Shelf: End-to-End Algorithmic Control of a Mobile Manipulator for Supermarket Stocking and Fronting
D. Peron, **V. Nan Fernandez-Ayala** and L. Segelmark
2025/2026 IEEE Conference on Automation Science and Engineering (**CASE**)

Enhancing Precision Agriculture Through Human-in-the-Loop Planning and Control
A. D. Shankar, P. Sujet, A. Matoses Gimenez, **V. Nan Fernandez-Ayala** et al.
2024 IEEE Conference on Automation Science and Engineering (**CASE**)

Multi-robot Human-in-the-loop Control under Spatiotemporal Specifications
Y. Zhang, **V. Nan Fernandez-Ayala** and D. V. Dimarogonas
2024 IEEE International Conference on Robotics and Automation (**ICRA**)

Distributed barrier function-enabled human-in-the-loop control for multi-robot systems
V. Nan Fernandez-Ayala, X. Tan and D. V. Dimarogonas
2023 IEEE International Conference on Robotics and Automation (**ICRA**)

Design of a High Altitude Long Endurance UAV for atmospheric imaging
V. Nan Fernandez-Ayala, L. Vimlati, A. Matoses Gimenez, H. Delmotte, M. Ivchenko and R. Mariani
2022 International Council of the Aeronautical Sciences (**ICAS**)

AWARDS, GRANTS & IMPACT

- Scholarship from Svensk-Spanska Stiftelsen, 2020–2022.
- Excellence Grant from Fundacion Botin, 2019–2020.
- UC3M International Mobility Grant for Georgia Tech exchange, 2018–2019.
- Excellence Grant from Fundacion Botin, 2017–2018.
- Two patents under review for vision-language model use for high-level planning in mobile manipulation at my startup Animum.
- Contributor and member in several popular public repositories for autonomous vehicles and embedded systems: PX4, ArduPilot and SimpleFOC.

ROBOTIC SYSTEMS

BlueROV2 underwater robots
ATMOS space platforms
Unitree G1 humanoid
Unitree Go2 quadruped
Hello Robot Stretch mobile manipulator
HEBI Rosie robotic platforms
Nexus mobile bases
Trossen WidowX robot arms
AgileX Piper robot arms
ALOHA2 robotic system
Crazyflie mini-quadrotors
Custom quadcopters and hexacopters

ENGINEERING

Open-source robotics: PX4 member and contributor for underwater and space vehicles, ArduPilot contributor for multicopters and SimpleFOC contributor for brushless motor controllers.

Robot infrastructure: Developed the widely used BlueROV2 system identification control and PX4 simulation tooling, plus sim2real and sim2sim workflows in Isaac Sim and Gazebo.

Mechatronics: Built multiple robotics projects, including my own CAN-HBC-S3, a Field-Oriented-Control BLDC motor controller, for a cable-driven robot arm design with custom reductors, using PCB design and prototyping in KiCad and embedded motor control on ESP32 platforms.

METHODS

Control and planning: visual servoing, system identification, MPC/MPPI controllers, safety-critical control using CBFs/PPC and task and motion planning with PDDL and behavior trees.

Learning and modeling: Koopman-based modeling, YOLO and VLM finetuning and deployment and OpenVLA development.

Formal and interactive autonomy: STL and LTL methods, digital twins in Unity and human-in-the-loop autonomy for both prediction and mixed-initiative control.

LEADERSHIP

Startup leadership: Co-Founder & CTO at Animum and technical advisor to startups for drone firefigting and manufacturing.

Research leadership: supervision of 9+ robotics theses and 15+ ongoing and completed publications at top robotics venues.

Teaching: teaching and lab assistant for master (EL2520) and bachelor (EL1010) courses in robotics, control and aerospace at KTH.